Vehicle & Sensor Estimation

Graduate Systems Engineering / Integration / Verification & Validation

PURPOSE

Estimate longitudinal speed and IMU bias from simulated acceleration and GPS speed.
Use the same run to explore recursive identification of road load parameters.

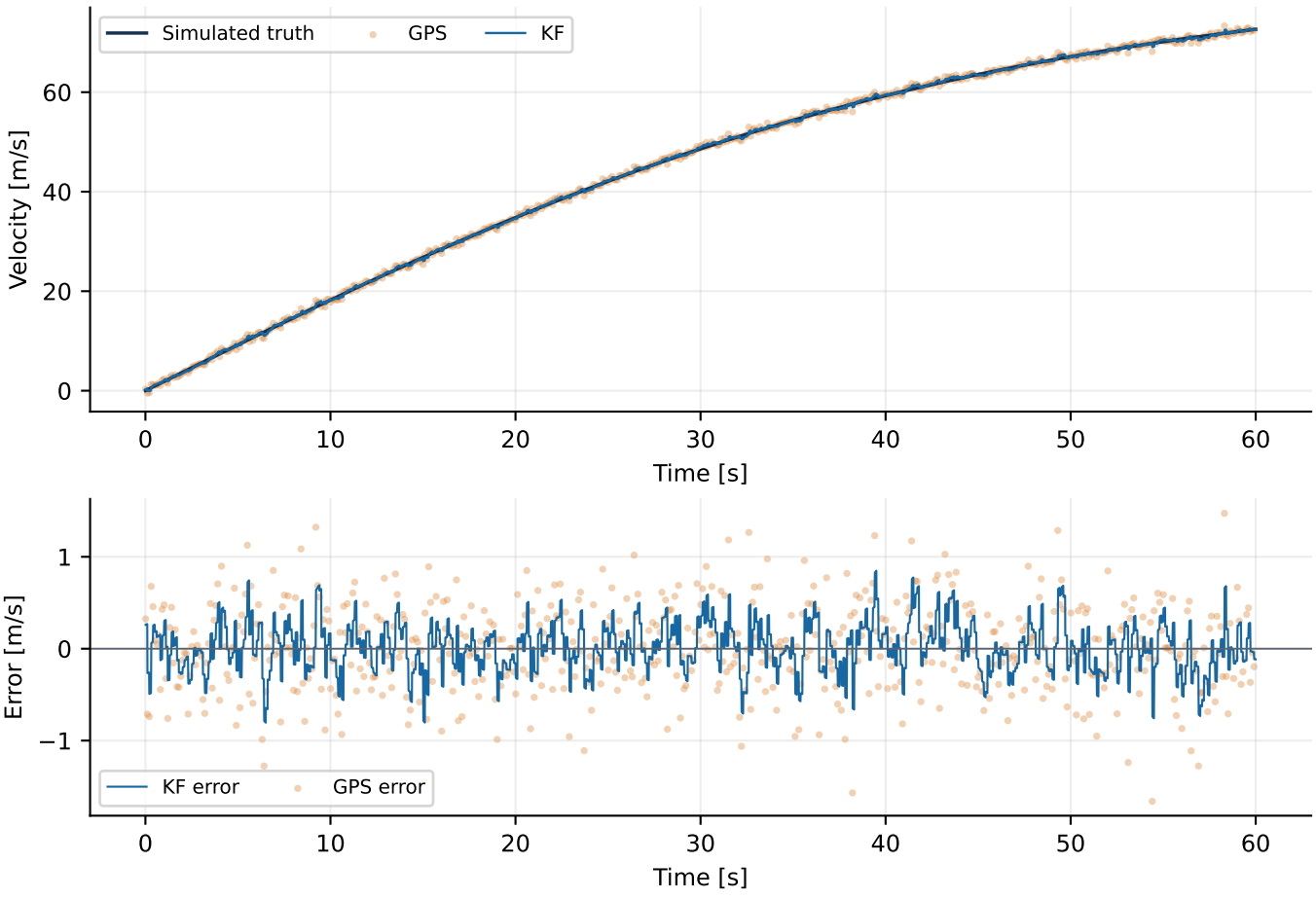
MY PROJECT WORK (AS REPRESENTED IN SOURCE)

- Force-balance vehicle model with aerodynamic drag and rolling resistance
- Noisy 100 Hz IMU and 10 Hz GPS simulation; two-state Kalman filter
- Recursive least squares (RLS) for CdA and rolling resistance coefficient

SYSTEM FLOW

Drive force → vehicle model → simulated truth → IMU / GPS → Kalman filter
↓
bias-corrected acceleration + speed → RLS

NOMINAL-RUN EVIDENCE



Verification Snapshot

Recomputed from results/nominal_results.npz generated with the supplied model

MEASUREMENT-CADENCE MATCHED RESULTS

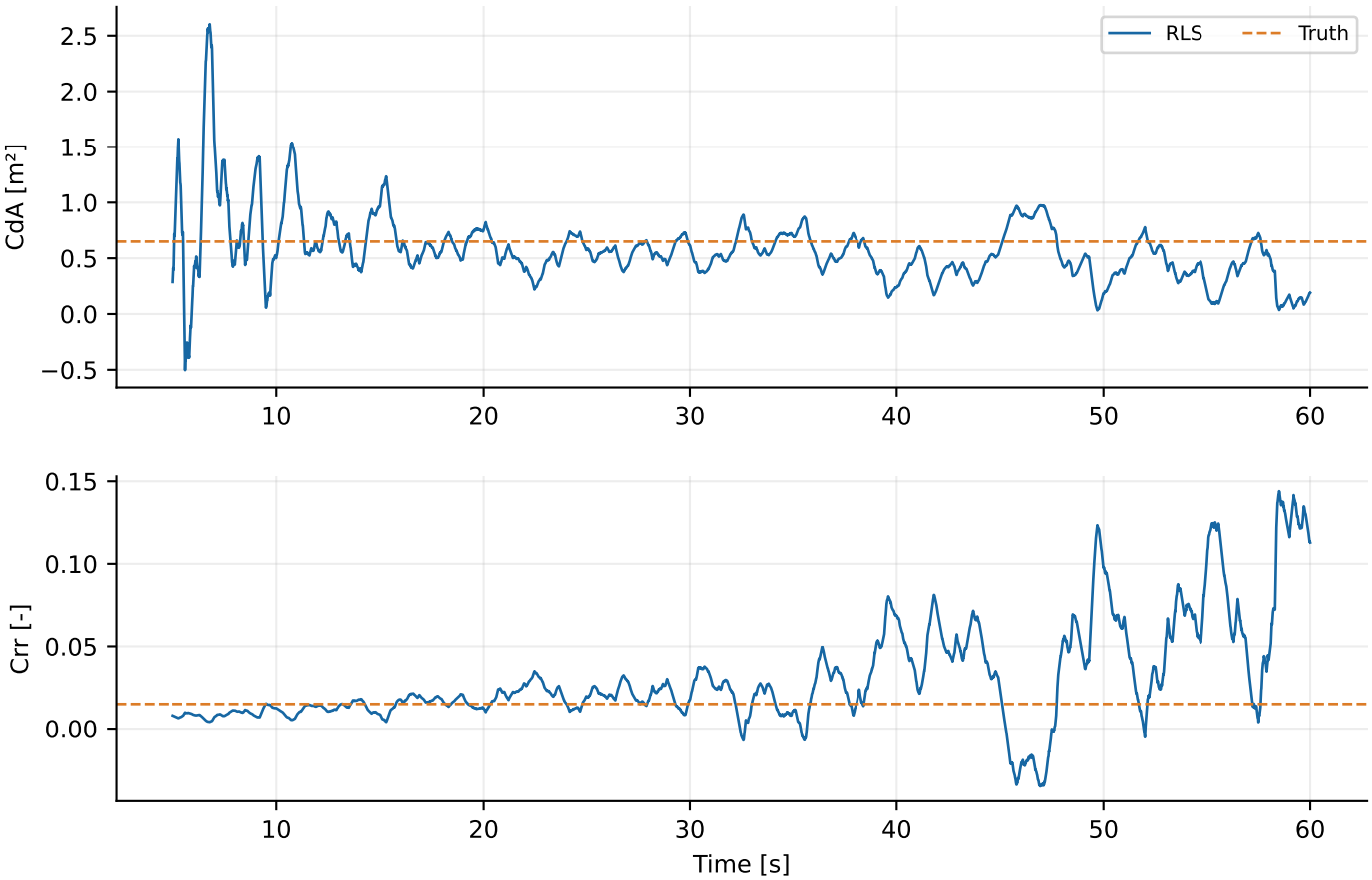
Velocity RMSE vs simulated truth at 600 GPS updates (0.01–59.91 s):

GPS	KALMAN FILTER
0.475 m/s	**0.272 m/s**

The KF value uses the state after each GPS update. Raw IMU integration at the same timestamps: 3.455 m/s RMSE; 100 Hz all-step KF RMSE: 0.275 m/s.

PARAMETER IDENTIFICATION: LIMITATION

Final CdA = 0.191 m² (truth 0.650); Crr = 0.113 (truth 0.015). Plot: 5–60 s.



WHAT THIS RUN DOES AND DOES NOT SHOW

KF has lower speed RMSE than GPS against simulated truth in this seeded run.
RLS parameters do not recover the known values; identification needs further study.
No real vehicle, hardware-in-loop, fault-injection or multi-seed validation is shown.

Method: seed 42; 60 s; 0.01 s step; constant 3000 N drive force.
Source, exact metrics and reproduction commands are included in the ZIP.